

Station Moves Matter: Documented Relocations and the Frankfurt Urban–Reference Temperature Gap, 1985–2025

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Abstract

We re-analyse the daily temperature difference between two Deutscher Wetterdienst (DWD) stations that DWD itself designates as an urban/reference pair, Frankfurt/Main-Westend and Frankfurt/Main (14,579 paired days, 1985–2025). The preregistered result reproduces exactly from freshly downloaded archives: a mean daily-mean gap of $+0.455^{\circ}\text{C}$ (95% block-bootstrap interval 0.432 to 0.478) and no detectable linear trend. A post-hoc check of two limitations the original report only mentioned shows the single average is not a stable property of the pair. The gap changed by -0.43°C (95% interval -0.54 to -0.29) across the urban station’s 2008 relocation and by $+0.19^{\circ}\text{C}$ (0.10 to 0.27) across the reference station’s 2014 relocation, each comparable to the whole mean gap, and our earlier statement that neither relocation produced an unusual jump was wrong. The daily-minimum gap ($+1.12^{\circ}\text{C}$) is about 2.5 times the daily-mean gap and the daily-maximum gap ($+0.12^{\circ}\text{C}$) is barely positive. A null trend is therefore uninformative while these steps remain unmodelled.

1 Question

How much warmer is urban Frankfurt than DWD’s own designated reference counterpart, with what uncertainty, and has the gap trended over four decades? We use daily mean temperature (TMK) at station 01424 (Frankfurt/Main-Westend, urban) and station 01420 (Frankfurt/Main, an airport that DWD labels the counterpart). The gap is the urban minus the reference value on days with both observations. This is a statement about two stations, not about the urban heat island of the whole city.

2 Preregistered analysis

The protocol fixed the stations, the period (1985-11-01 to 2025-12-31), the rule for missing days, a 30-day moving-block bootstrap for the mean gap (daily gaps are strongly autocorrelated), and an ordinary least-squares trend on annual mean gaps for the 40 years with at least 300 valid paired days. A post-release Newey–West (three-lag) interval [1] was added as a sensitivity because the annual residuals have lag-1 correlation 0.657. The annual-mean trend is -0.0031°C per year (classical 95% interval -0.0071 to $+0.0008$, $p = 0.118$; Newey–West -0.0079 to $+0.0016$, $p = 0.186$): no linear trend is detected. All of this regenerates identically from a fresh download of the two DWD archives.

Period	n days	Mean gap ($^{\circ}\text{C}$)	95% interval
Full period	14,579	+0.455	0.432 to 0.478
Winter (DJF)	3,641	+0.489	0.453 to 0.522
Spring (MAM)	3,619	+0.541	0.497 to 0.589
Summer (JJA)	3,649	+0.415	0.361 to 0.471
Autumn (SON)	3,670	+0.374	0.341 to 0.406

Table 1: Preregistered mean gap (daily mean temperature) with block-bootstrap intervals.

3 Post-hoc extensions

The original report named two untested limitations: the daily minimum and maximum (TNK, TXK) might show different gaps, and the documented relocations (urban 2008-07-01, reference 2014-10-22) might introduce steps. We tested both with the same block bootstrap; none of this was preregistered and every result is reported. By segment the daily-mean gap is +0.506 (1985–

Variable	Mean gap ($^{\circ}\text{C}$)	95% interval	Newey–West trend ($^{\circ}\text{C}/\text{yr}$)
Daily mean (TMK)	+0.455	0.432 to 0.477	−0.0031 (−0.0079 to 0.0016)
Daily minimum (TNK)	+1.120	1.086 to 1.153	+0.0014 (−0.0077 to 0.0106)
Daily maximum (TXK)	+0.120	0.098 to 0.143	−0.0029 (−0.0069 to 0.0010)

Table 2: The gap by variable. Spring has the largest gap for all three.

After – before (3-year windows)	TMK	TNK	TXK
Urban move, 2008-07-01	−0.433 (−0.544, −0.292)	−0.271 (−0.386, −0.132)	−0.241 (−0.370, −0.120)
Reference move, 2014-10-22	+0.187 (0.103, 0.267)	+0.414 (0.234, 0.595)	+0.101 (0.026, 0.173)

Table 3: Step in the mean gap across each documented relocation, with block-bootstrap 95% intervals.

2008-06), +0.337 (2008-07 to 2014-10) and +0.417 (2014-10 onward). The annual means make the step visible: 0.72, 0.82 and 0.81 $^{\circ}\text{C}$ in 2005–2007 against 0.36 and 0.27 $^{\circ}\text{C}$ in 2009–2010. An annual regression with level shifts at 2009 and 2015 and Newey–West errors cannot separate steps from trend with only 40 points (slope +0.0019, 95% interval −0.0147 to +0.0185; 2009 step −0.188, −0.485 to +0.108), so the direct window comparison is the more precise instrument.

4 Discussion

The documented 2008 move was described as “slight” and within the same district, yet the gap fell by about as much as its entire full-period mean. The full-period mean therefore averages over three different station histories, and the null trend result, which a reader could take as “the urban heat island is not changing”, says little about real change while instrument and siting steps of this size are present. Two caveats apply: three-year windows also contain real interannual variability and any true change in the surroundings, so the steps are coincident with, not proven caused by, the moves; and the annual regression’s wide intervals show the data alone cannot homogenise the series. Physically the pattern is coherent with the urban heat island being mainly a night-time phenomenon, with a gap in the daily minimum 2.5 times that of the daily mean. A proper estimate

of change would need the stations' homogenisation metadata or a reference network, which this study does not have.

5 Correction

The earlier report stated that neither relocation coincides with an unusual jump in the annual-mean-gap series. The annual series contradicts that. The statement is corrected in the repository (erratum in the report); no frozen registry value changed.

6 Limitations

Two stations and one pairing; the reference is an airport, not a rural site; elevation and distance (15.4 km) differ; only three temperature variables; the three-year window is an arbitrary choice fixed in the code before the step results were seen; and the results are descriptive, not a homogenisation.

Reproducibility and AI assistance

Code, frozen registry, post-hoc extensions and tests:

<https://github.com/lindgreendavid/climate-twin-frankfurt>

Interactive laboratory:

<https://climate-twin-frankfurt-interactive.lindgreendavid.workers.dev>

The analysis code, tests and drafts of this text were produced with substantial assistance from an AI system (Claude, Anthropic); the author is responsible for the content.

References

- [1] W. K. Newey and K. D. West, A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix, *Econometrica* **55**, 703–708 (1987).